

Module 2

Q. Project selection models

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1. Project selection models are methods used by organizations to decide which projects should be started or approved.
2. These models help managers choose projects that best support the company's goals and available resources.
3. Project selection models help in comparing multiple project proposals in a systematic way.
4. These models reduce personal bias and help in making objective decisions.
5. Project selection models ensure that selected projects provide maximum benefit to the organization.
6. Organizations often receive many project ideas, but only a few can be selected due to limited budget, time, and manpower.
7. Project selection models help managers evaluate the feasibility and value of different projects.
8. These models are mainly divided into two categories: **Numeric / Scoring Models** and **Non-Numeric Models**.

1. Numeric / Scoring Models

1. Numeric models are project selection methods that use numbers or mathematical calculations to evaluate projects.
2. These models assign scores or financial values to each project.
3. Projects with higher scores or better financial returns are usually selected.
4. Numeric models help in comparing projects in a quantitative and objective manner.
5. These models are widely used in organizations because they provide measurable results.
6. Numeric models are useful when the organization wants to evaluate projects based on cost, profit, risk, and benefits.

Scoring Model

1. The scoring model is a commonly used numeric model for project selection.
2. In this model, different criteria are identified for evaluating projects.
3. Each criterion is given a weight according to its importance.
4. Every project is scored based on how well it meets each criterion.
5. The final score of the project is calculated by multiplying the score with the weight.
6. The project with the highest total score is usually selected.
7. This model helps in considering multiple factors at the same time.
8. The scoring model is flexible because organizations can choose their own evaluation criteria.

Example of Scoring Model

1. A company may evaluate projects based on criteria such as cost, expected profit, risk level, and strategic importance.
2. Suppose the company gives **profit a weight of 40%, cost a weight of 30%, risk a weight of 20%, and strategic importance a weight of 10%.**
3. Each project is then scored on these criteria.
4. The total weighted score is calculated to determine the best project.
5. For example, if **Project A receives the highest total score**, it will be selected over other projects.

2. Non-Numeric Models

1. Non-numeric models are project selection methods that do not rely on mathematical calculations.
2. These models are based on judgment, experience, or strategic considerations.
3. Non-numeric models are useful when quantitative data is not available.
4. These models help organizations select projects that align with long-term goals.
5. Non-numeric models are often used during the early stages of project selection.

Types of Non-Numeric Models

Sacred Cow Model

1. In the Sacred Cow model, a project is suggested by a senior executive or top manager.
2. The project is accepted mainly because of the authority of the person proposing it.
3. The project is rarely questioned by other employees.
4. This model is commonly seen in organizations with strong leadership influence.
5. For example, if the CEO suggests developing a new mobile application, the company may approve the project immediately.

Operating Necessity Model

1. In the Operating Necessity model, a project is selected because it is essential for the organization's survival.
2. The project is necessary to continue normal operations.
3. These projects are usually urgent and unavoidable.
4. For example, replacing a damaged factory machine is necessary to keep production running.

Competitive Necessity Model

1. In the Competitive Necessity model, a project is selected to maintain the company's position in the market.
2. The project helps the organization stay competitive with other companies.

3. For example, if competitors launch an online shopping platform, a retail company may start its own e-commerce project.

Product Line Extension Model

1. In this model, projects are selected to expand the company's existing products or services.
2. The goal is to increase market share or attract new customers.
3. For example, a smartphone company may launch a new variant of its existing phone model.

Q. Explain how a project is started in an organization. Also describe the concept of an effective project team and the stages of team development.

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1. How to Get a Project Started

1. A project is started when an organization identifies a need or problem that must be solved through a planned activity.
2. The first step in starting a project is identifying a clear **goal or objective**.
3. The project goal should explain what the project wants to achieve.
4. The objective should be specific, measurable, and achievable.
5. For example, a company may start a project to **develop a new mobile application for online shopping**.

Identifying the Need for the Project

6. A project usually begins when an organization identifies a business opportunity or problem.
7. The project idea may come from customers, employees, or market demand.
8. For example, a company may start a project to create a **digital payment system** because customers prefer online transactions.

Conducting Feasibility Study

9. A feasibility study is conducted to check whether the project is practical and achievable.
10. This study evaluates the **technical, financial, and operational feasibility** of the project.
11. Technical feasibility checks whether the required technology and skills are available.
12. Financial feasibility checks whether the organization can afford the project cost.
13. Operational feasibility checks whether the project can be successfully implemented in the organization.

Preparing the Project Proposal

14. A project proposal is a document that explains the purpose and scope of the project.
15. The proposal includes project objectives, expected benefits, estimated cost, and timeline.
16. The proposal helps management understand the importance of the project.
17. For example, a proposal for a **college management system** may include features like student records, attendance tracking, and fee management.

Project Approval

18. After the proposal is prepared, it is presented to senior management or stakeholders.
19. The stakeholders evaluate whether the project aligns with the organization's goals.
20. If the project is beneficial and feasible, it receives approval.

21. Once approved, the project officially begins.

Assigning a Project Manager

22. A project manager is appointed to lead and manage the project.

23. The project manager is responsible for planning, organizing, and controlling the project activities.

24. The project manager also coordinates with team members and stakeholders.

25. For example, in a **software development project**, the project manager manages developers, designers, and testers.

2. Effective Project Team

1. An effective project team is a group of individuals who work together to achieve the project goals.
2. The team members should have different skills and expertise required for the project.
3. An effective team improves productivity and project success.
4. Good communication and cooperation are important for an effective team.

Characteristics of an Effective Project Team

5. An effective team has a **clear understanding of project goals**.
6. Each team member understands their roles and responsibilities.
7. Team members communicate openly and share information.
8. Team members respect each other's ideas and suggestions.
9. The team leader provides proper guidance and support.
10. The team works collaboratively to solve problems.

Importance of an Effective Project Team

11. An effective project team helps in completing the project on time.
12. The team improves the quality of project results.
13. Good teamwork reduces conflicts and misunderstandings.
14. Effective teams increase creativity and innovation.
15. For example, in a **website development project**, designers, programmers, and testers must work together to create a successful website.

3. Stages of Team Development

1. Team development refers to the process through which a group of individuals becomes an effective working team.
2. Teams usually go through several stages before becoming fully productive.
3. The most common model of team development was proposed by **Bruce Tuckman**.
4. The model includes five stages: **Forming, Storming, Norming, Performing, and Adjourning**.

Forming Stage

5. The forming stage is the first stage of team development.
6. In this stage, team members meet each other and learn about the project.
7. Team members are usually polite and cautious.
8. Roles and responsibilities are not fully clear during this stage.
9. The team leader provides guidance and direction.
10. For example, when a new **software development team** is created, members introduce themselves and discuss the project goals.

Storming Stage

11. The storming stage occurs when team members start expressing their opinions.
12. Conflicts or disagreements may arise during this stage.
13. Team members may compete for leadership or influence.
14. Differences in ideas, working styles, or responsibilities may cause tension.
15. Proper communication and leadership help resolve conflicts.

Norming Stage

16. In the norming stage, team members start working together effectively.
17. Team members develop trust and mutual understanding.
18. Rules and working procedures are established.
19. The team begins to cooperate and support each other.
20. Productivity improves during this stage.

Performing Stage

21. The performing stage is when the team becomes highly productive.
22. Team members work efficiently toward achieving the project goals.
23. The team is able to solve problems independently.
24. Cooperation and coordination are strong during this stage.
25. For example, a **product development team** successfully completes tasks such as coding, testing, and deployment.

Adjourning Stage

26. The adjourning stage occurs when the project is completed.
27. The team finishes its work and prepares the final results.
28. Team members may move to other projects or responsibilities.
29. The team may review the project performance and lessons learned.
30. For example, after completing a **mobile app project**, the team may conduct a final meeting to evaluate the results.

Q. Stages / Initiation of Projects

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1. Project initiation is the first stage of the project management process.
2. In this stage, the idea of the project is identified and evaluated.
3. Project initiation helps determine whether the project should be started or not.
4. The main goal of this stage is to define the project purpose, objectives, and feasibility.
5. Proper initiation helps avoid risks and ensures that the project aligns with organizational goals.
6. For example, a company may initiate a project to **develop a new mobile application for customers**.

1. Identifying the Project Idea

7. The first step in project initiation is identifying a project idea or opportunity.
8. The project idea may arise due to a business problem, customer demand, or technological advancement.
9. Organizations often analyze market trends to identify new project opportunities.
10. For example, a bank may start a project to create a **mobile banking application** because customers prefer online banking services.

2. Defining Project Objectives

11. After identifying the idea, clear project objectives are defined.
12. Project objectives describe what the project aims to achieve.
13. Objectives help guide the project team in the right direction.
14. Objectives should be clear, realistic, and measurable.
15. For example, the objective of an e-commerce project may be **to build an online shopping website within six months**.

3. Conducting a Feasibility Study

16. A feasibility study is conducted to determine whether the project is practical and achievable.
17. This study analyzes different aspects such as technical feasibility, financial feasibility, and operational feasibility.
18. Technical feasibility checks whether the required technology and expertise are available.
19. Financial feasibility checks whether the project can be completed within the available budget.
20. Operational feasibility checks whether the organization can successfully implement the project.

4. Identifying Stakeholders

21. Stakeholders are individuals or groups who are affected by the project.
22. Stakeholders may include managers, employees, customers, investors, and suppliers.
23. Identifying stakeholders helps understand their needs and expectations.

24. Stakeholder involvement improves project support and communication.

25. For example, in a **college management system project**, stakeholders may include students, teachers, and administrators.

5. Preparing the Project Proposal

26. A project proposal is prepared to explain the project idea and plan.

27. The proposal includes project objectives, scope, estimated cost, and expected benefits.

28. The proposal helps decision makers understand the value of the project.

29. A well-prepared proposal increases the chances of project approval.

30. For example, a proposal for a **smart parking system** may explain how it will reduce traffic congestion.

6. Project Approval

31. The project proposal is reviewed by senior management or sponsors.

32. Management evaluates whether the project aligns with organizational goals.

33. The project is approved if it is beneficial and feasible.

34. Once approved, the project moves to the planning stage.

35. For example, a company may approve a project to **develop an AI-based customer support chatbot**.

7. Assigning the Project Manager

36. After approval, a project manager is appointed to lead the project.

37. The project manager is responsible for planning, organizing, and managing project activities.

38. The project manager coordinates with team members and stakeholders.

39. The project manager ensures that the project stays within scope, time, and budget.

40. For example, in a **software development project**, the project manager manages developers, testers, and designers.

8. Creating the Project Charter

41. A project charter is a formal document that officially starts the project.

42. The project charter defines the project purpose, objectives, and stakeholders.

43. The charter also gives authority to the project manager to manage the project.

44. The document acts as a reference throughout the project lifecycle.

45. For example, a project charter for a **website development project** may define project scope, timeline, and responsibilities.

Q. Steps to Start a Project

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1. Starting a project means beginning a planned activity to achieve a specific goal or objective.
2. The process of starting a project helps define what the project will do and how it will be completed.
3. Proper project start-up ensures that the project is clear, organized, and achievable.
4. The starting stage of a project is also called the **project initiation phase**.
5. This stage helps the organization understand the purpose and benefits of the project.
6. For example, a company may start a project to **develop a new online shopping website**.

1. Defining the Project Objective

7. The first step in starting a project is defining clear project objectives.
8. Project objectives describe what the project aims to achieve.
9. Objectives help guide the project team in the right direction.
10. Objectives should be clear, specific, and measurable.
11. For example, the objective of a software project may be **to create a mobile application within six months**.

2. Identifying the Problem or Opportunity

12. A project is usually started to solve a problem or take advantage of an opportunity.
13. Organizations analyze market conditions and customer needs before starting a project.
14. This step helps identify why the project is important.
15. For example, a bank may start a project to develop a **digital payment system** because customers prefer online transactions.

3. Conducting Feasibility Study

16. A feasibility study is conducted to determine whether the project is practical and achievable.
17. The feasibility study checks whether the project can be completed with available resources.
18. Technical feasibility checks whether the required technology and skills are available.
19. Financial feasibility checks whether the project is affordable.
20. Operational feasibility checks whether the organization can successfully implement the project.

4. Identifying Stakeholders

21. Stakeholders are individuals or groups who are affected by the project.
22. Stakeholders may include managers, employees, customers, and investors.
23. Identifying stakeholders helps understand their needs and expectations.
24. Proper communication with stakeholders improves project support.

25. For example, in a **college management system project**, stakeholders include students, teachers, and administrators.

5. Defining the Project Scope

26. Project scope defines the boundaries and limitations of the project.

27. Scope explains what work will be included and what will not be included in the project.

28. A clear scope prevents confusion and unnecessary work.

29. Scope helps the project team focus on the required tasks.

30. For example, the scope of a **website project** may include product listing, payment system, and order tracking.

6. Preparing the Project Proposal

31. A project proposal is prepared to explain the project plan.

32. The proposal includes project objectives, scope, estimated cost, and expected benefits.

33. The proposal helps management understand the value of the project.

34. A well-prepared proposal increases the chances of project approval.

35. For example, a proposal for a **smart parking system** may explain how it will reduce traffic congestion.

7. Getting Project Approval

36. The project proposal is presented to senior management or project sponsors.

37. Management reviews the proposal and decides whether to approve the project.

38. The project is approved if it is beneficial and feasible.

39. Once approved, the project officially begins.

40. For example, a company may approve a project to **develop an AI-based chatbot for customer service**.

8. Appointing the Project Manager

41. After approval, a project manager is appointed to lead the project.

42. The project manager is responsible for planning, organizing, and controlling the project activities.

43. The project manager coordinates with team members and stakeholders.

44. The project manager ensures that the project is completed within time and budget.

45. For example, in a **software development project**, the project manager manages developers, testers, and designers.